

ARTI403: Image Processing

Lab 1: Basics of Programming with Python

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Objective

To explain how digital images are represented and manipulated in a computer.

Tools

Python 3, Jupyter, OpenCV, NumPy, Pillow, Matplotlib.

Procedural Steps

Steps 1 to 4: Setup

I started Jupyter and installed the libraries I need for the course: OpenCV, scikit-image, Pillow, Matplotlib and NumPy.

Task 1: Exercise with Spyder

I practiced the Python Basics and NumPy cheat sheets.

Task 2: Loading and Visualizing Images

I loaded cameraman.tif with OpenCV (it becomes a NumPy array) and displayed it with Matplotlib. Then I loaded lena_gray_256.tif with PIL and displayed it.

```
img = cv2.imread('cameraman.tif', cv2.IMREAD_GRAYSCALE)
plt.imshow(img, cmap='gray')
```

```
img2 = Image.open('lena_gray_256.tif')
plt.imshow(img2, cmap=cm.Greys_r)
```

Camerman Image - OpenCV



Figure 1: Camerman image (OpenCV)

Grayscale Image - PIL



Figure 2: Grayscale image (PIL)

Task 3: Image Storing

I saved the OpenCV image as new_image.jpg and the PIL image as new_image2.jpg.

```
cv2.imwrite('new_image.jpg', img)
img2.save('new_image2.jpg')
```

Task 4: Display Image as an Array

The OpenCV image is already a NumPy array, so I printed its shape and values. For the PIL image I converted it with np.array() first.

```
print(img.shape)           # (512, 512)
print(np.array(img2).shape) # (256, 256)
```

Both images are 8-bit (uint8), so the pixel values are from 0 to 255.

Assessment: NumPy Operations

I used NumPy on the array of the Lena image. These are the results:

```
Minimum intensity: 0
Maximum intensity: 255
Mean intensity: 129.07
Standard deviation: 73.05
```

I also made a negative image ($255 - \text{image}$), a brighter image (+50), a crop of the image, and a horizontal flip.



Figure 3: Negative, brighter, cropped and flipped images